PRODUCT DIRECTION / OCTOBER 2026

TankLook

Development priorities & roadmap

Build 560 • Based on the supplied roster and previously reviewed model sheets

A visual sandbox for arranging realistically scaled living things and objects inside transparent enclosures—freshwater, marine, terrestrial, mixed, or unconventional.



Illustrative enclosure concepts: freshwater aquarium, arboreal vivarium, paludarium, arid terrarium, jumping-spider enclosure, and coral reef.

Executive recommendation

Prioritize capabilities that unlock entire enclosure categories before continuing broad species expansion. First, make dry terrestrial scenes convincing. Then add surface-aware placement and the initial reptile/invertebrate body plans. Continue marine and freshwater coverage selectively.

Product rule — One shared placement and visualization system, not a separate editor for each ecosystem.

Primary success metric — Can a user create a convincing representative scene quickly, at realistic scale, without workarounds?

All illustrations in this document are AI-generated conceptual references, not screenshots of TankLook or validated animal-model renders.

1 | Ranked development priorities

The roster is already strong for freshwater community aquariums. The highest-value gaps are in dry habitats, terrestrial life, mixed land/water composition, and the tools needed to arrange believable scenes.

Rank	Addition	Value	Est. effort
1	Dry terrestrial substrate and land regions	Very high	Medium–high
2	Surface-aware placement; depth and occlusion	Very high	Medium–high
3	Terrestrial plants, wood, and habitat materials	Very high	Medium
4	Reusable animal anatomy and pose systems	Very high	High
5	First reptiles and terrestrial invertebrates	High	High
6	Bulk placement, grouping, and natural variation	High	Medium
7	Custom 2D asset import	High	Medium
8	Enclosure presentation and structural customization	High	Medium
9	Expanded marine life and coral forms	Medium–high	Medium
10	Additional freshwater fish and variants	Medium	Low–medium

Why this order?

One new environment capability may serve dozens of species and asset types. By contrast, a new tetra adds little if the application already supports similar silhouettes, sizes, and swimming positions. Similarly, a new reptile is less useful if it cannot sit on a branch or stand naturally on the ground.

Current inventory: build 560

Library area	Entries	Assessment
Freshwater fish	34	Good coverage of common community fish
Freshwater invertebrates	6	Useful starter set
Saltwater fish / inverts	9 / 3	Functional starting point; limited variety
Amphibians	6	Existing bridge into terrestrial scenes
Reptiles / terrestrial inverts	0 / 0	Entirely missing categories
Plants / corals	19 / 4	Aquatic bias; coral groups generic
Stones / wood / caves	6 / 4 / 3	Good basics; limited dry-habitat coverage

Effort ratings are directional architectural estimates. Feature completion was not verified against the live editor.

2 | Environment capabilities

The fundamental expansion is from “an underwater world” to “a glass container that can hold any combination of air, water, terrain, and living things.” Avoid exclusive aquarium/terrarium modes: control the environment through its properties.



Mixed land/water composition



Elevated and vertical structure

Feature	Priority	Implementation direction
Dry tank; zero water level	P0	Keep all placement tools functional without water.
Terrestrial substrates	P0	Soil, sand, clay, bark, leaf litter, and layered ground.
Substrate thickness	P0	Editable depth with visible cross-section.
Land/water boundaries	P1	Simple editable regions; defer unrestricted terrain sculpting.
Elevated surfaces	P1	Branches, stones, shelves, and ledges serve as anchors.
Textured backgrounds	P1	Cork panels, rock walls, botanical surfaces.
Portrait/landscape enclosure	P1	Arbitrary rectangular proportions with correct viewport.
Irregular terrain	P2	Slopes, steps, banks, and raised land masses.

Architecture principle

A dry terrarium is a scene with no water; a paludarium is a scene with a partial water volume. They should use the same object, scale, placement, grouping, and rendering machinery. Do not introduce an ecosystem selector that arbitrarily disallows objects.

3 | First terrestrial asset batch

Build a small, versatile environment kit before producing many terrestrial animals. These assets immediately enhance the existing dart frog, White's tree frog, and fire-bellied toad while supporting later reptile/invertebrate additions.

Asset	Priority	Core visual purpose
Soil / coco fiber	P0	Foundational planted-terrarium substrate
Sand	P0	Arid ground and contrasting surface texture
Cork bark flat / tube	P0	Backgrounds, shelters, climbing volumes
Leaf litter	P0	Forest floor and natural density
Pothos	P0	Large trailing and climbing foliage
Bromeliad	P0	Distinct tropical silhouette
Tropical fern	P0	Fine-structured midground mass
Fittonia	P1	Compact patterned foreground plant
Sphagnum moss	P1	Mossy coverage and surface texture
Flexible vine	P1	Connects branches and vertical features
Rock ledge / slab	P1	Stone shelves and arid scene structure
Terrestrial hide	P1	Reptile and invertebrate microhabitats
Climbing branch	P1	Natural supports distinct from submerged driftwood

Modeling ROI

Do not individually model every leaf or clump. Use reusable foliage forms plus scattering/instancing for moss, leaf litter, fine grass, and substrate textures. These techniques create density without multiplying library-management costs.



Tropical surface coverage



Dry textures, stones, and shelters

4 | First terrestrial animal models

Select animals that introduce genuinely different body plans and positioning requirements—not merely five popular names.



Crested gecko · arboreal climbing



Leopard gecko · terrestrial stance



Regal jumping spider · fine appendages



Ball python · coiled / extended body



Tarantula · articulated legs and hair

Animal	Pipeline capability established
Crested gecko	Lizard anatomy; gripping toe pads; branch/wall poses
Leopard gecko	Ground-contact stance; tail and limb proportions
Regal jumping spider	Small-scale arthropod anatomy and leg segmentation
Ball python	Serpentine bodies; coiled and extended pose families
Tarantula	Large arachnid proportions; joint chains and hair coverage

Pipeline prerequisite

Earlier model-sheet reviews found recurring structural issues in shrimp, snail shells, and species-defining fish fins. Those findings came from model v1 (build 217), not build 560; verify whether they remain before using the same generators to scale into reptiles and terrestrial invertebrates.

Fish can appear natural in an arbitrary suspended pose. Land animals usually cannot: a gecko must visibly cling to a surface, and a leopard gecko needs a convincing ground-contact stance. Pose and anchors are core visual requirements, not later polish.

5 | Composition tools and unlimited variation

TankLook is an arrangement application. Efficient composition is potentially more valuable than adding dozens of additional closely related species.

Feature	Priority	User-visible result
Multi-select and duplication	P0	Replicate fish schools, repeated plants, or rock clusters.
Brush / scatter placement	P0	Quickly populate litter, moss, pebbles, and carpeting plants.
Controlled random variation	P0	Small changes in angle, scale, and spacing to avoid repetition.
Surface anchors	P0	Attach animals/plants to floor, glass, wood, stone, or wall.
Depth and occlusion	P0	Correct foreground/background order, overlap, and contact.
Pose variants	P1	Perching, climbing, resting, swimming, and coiling poses.
Reusable object groups	P1	Move, rotate, copy, and save multi-object arrangements.
Custom asset import	P1	Import transparent PNGs with real-world length and depth placement.
Appearance overrides	P2	Color and dimensional customization where desired.

Custom assets: the escape hatch

No finite official species library can cover everything. A transparent PNG import with a user-specified real-world size could instantly support unusual organisms or decorations. Full 3D imports can come later; both should honor the same coordinate and scaling system.

Modeling quality as shared infrastructure

Body plans — Use separate anatomical templates rather than adapting a generic fish-like shape.

Attachments — Parameterize fin bases, joints, legs, shell openings, eyes, mouths, and other structural landmarks.

Pose validation — Test front/side/top silhouettes and representative mounted poses before production release.

Lighting and materials — Evaluate transparent fins, hair, shells, and other materials in bright preview and tank lighting.

6 | Selective aquatic expansion

Continue freshwater and reef development, but target missing visual types: body plans, growth forms, scale classes, and structural diversity.



Corals and marine forms remain underrepresented



Freshwater: favor different silhouettes and scales

Marine priorities

Addition	Priority	What it adds visually
Zoanthid colony	P1	Dense, low polyp mats
Hammer coral	P1	Fleshy branching or wall formation
Bubble-tip anemone	P1	Tentacular non-coral focal point
Torch coral	P1	Long projecting tentacles
Acropora	P1	Fine-branching stony coral
Plating Montipora	P2	Horizontal shelves and plates
Sea fan / gorgonian	P2	Vertical branching plane
Macroalgae	P2	Marine planted-display aesthetics
Blue tang / mandarin dragonet	P2	Distinctive large and bottom-oriented fish

Freshwater priorities

Addition	Priority	Coverage gap addressed
Pygmy corydoras	P1	Miniature bottom-dweller scale
Hillstream loach	P1	Flattened substrate-hugging silhouette
Yellow lab cichlid	P1	First African mbuna
Boesemani rainbowfish	P1	Rainbowfish body form and larger active fish
Cockatoo apistogramma	P2	Dwarf cichlid and distinct fins
White cloud mountain minnow	P2	Cool-water schooler
Comet goldfish	P2	Long-bodied goldfish distinct from fancy varieties

Plant follow-ons: Monte Carlo, dwarf hairgrass, hornwort, and Anacharis where existing generic assets do not already capture their growth forms.

7 | The enclosure itself

Structural customization expands presentation without requiring a different editor. Prioritize the common enclosure shapes and elements that visibly change the scene.

Feature	Priority	Why it matters
Tall / horizontal rectangular presets	P1	Quick layouts for arboreal vs. terrestrial displays
Background panels and wall textures	P1	Large effect on the visible finished environment
Mesh lids and front doors	P2	Recognizable terrarium construction
Equipment models	P2	Lights, filters, heaters, mister lines, feeding dishes
Lighting direction and intensity	P2	Scene mood, shadows, and presentation
Cylinders and bowls	P3	Unconventional glass display shapes
Custom irregular enclosure geometry	P3	Broad flexibility with high implementation cost

What not to build yet

Defer arbitrary 3D enclosure geometry and full animal motion/behavior simulation until the static placement system convincingly handles rectangular aquarium, terrarium, paludarium, and reef displays.

8 | Four release milestones

Release 1 — Convincing dry enclosure

Dry substrates, terrestrial texture layers, wall materials, and a dozen starter plants/hardscape assets.

Acceptance test: A believable planted glass terrarium with no animal whatsoever.

Release 2 — Terrestrial animals

Surface anchoring, appropriate static poses, and five representative terrestrial species.

Acceptance test: Build crested gecko, leopard gecko, and jumping-spider setups without placement hacks.

Release 3 — Unified land and water

Partial-water regions, raised banks, emergent plants, and correct occlusion across the interface.

Acceptance test: Build a believable mixed land/water paludarium with branches, aquatic planting and frogs.

Release 4 — Diversity and freedom

Marine forms, selective freshwater species, efficient duplication/scattering, and initial custom import.

Acceptance test: Build reef, freshwater, terrestrial and mixed displays from the same editor.

9 | Quality benchmarks and decisions

Use a small set of finished example scenes as the completion metric. These measure whether the editor supports attractive, recognizable arrangements at realistic scale, with minimal workarounds.

Benchmark scene	Capabilities under test
20-gallon planted community aquarium	Scale; schooling groups; foliage density; visual depth
40-gallon reef aquarium	Coral forms; rockwork; saltwater invertebrates
18 × 18 × 24 in arboreal vivarium	Vertical structures; branches; climbing poses
20-gallon-long leopard gecko terrarium	Dry substrate; hides; stones; floor contact
Jumping spider microhabitat	Small-enclosure scale; fine appendages; close-up view
Mixed paludarium	Land/water boundary; emergent plants; amphibians
Closed botanical terrarium	Dense foliage; layered textures; plant-only composition

Final recommendation

Make a fully convincing planted terrestrial enclosure the next major milestone, then add the first reptile and terrestrial-invertebrate body plans. Keep the composition editor ecosystem-neutral. Add aquatic species selectively where they introduce distinctly new silhouettes or habitats.

Decisions to confirm against the live build

- Which dry substrate, surface, and water-level capabilities already exist?
- Does object depth and occlusion work consistently across every asset class?
- Can fish, plants, and decorations be grouped, cloned, scattered, and rotated?
- Can 3D animal meshes attach to floor, rock, branch, or background planes?
- Which model-build-217 audit defects have been resolved in build 560?

Planning basis: user-supplied build 560 roster and TankLook model anatomy audit dated 2026-10-14. Illustrations are generated concepts, not photographs or captures of the actual application. Effort estimates are provisional.